The Islamia university of Bahawalpur

Faculty of computer science

Department of (IT)



(Final year project proposal)

Campus lost & found system with AI image matching

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Project title: campus lost &found with AI image matching:

1. project Abstract

Campus lost &found is secure mobile and web platform designed to automate the tracking and recovery of misplaced items on campus. Built using **Flutter, Google ML Kit,Node.js/Express and Mongo DB Atlas,**the app uses on-device AI to auto tag found items(object type and color)from photos and cross matches them with lost The system enforce strict university email authentication **(@iub.edu.pk)** protects students pravicy through in app prevents fake claims using OTP hand over validation,logs detailed lifecycle timestamps and supports optional finder reward/commission model at zero operational hosting cost.

2 . Problem statement

Existing campus lost & found method rely on manual social media posts or physical notice boards. These methods suffer from:

1. Unverified claims and fraud:

   Anyone can fake ownership without proper verification.

2. Lack of privacy:

   Students are forced to share their phone number on social groups.

3. Manual inefficient searching:

   Owners must scroll through hundreds of posts manually.

4. No audit trail or incentives:

   No record when/where items are returned and no structured incentive for honest finder.

3. Key objectives

1. AI auto tagging:

   Automatically classify object category and primary color on device using **Google ML Kit** ($0 cloud API fee).

2. Automated Matching:

   Execute a weighted search query (category, color , location, time) and trigger instant firebase cloud Messaging **(FCM)**alerts.

3. Domain security:

   Enforce **@iub.edu.pk** email verification to restrict app access strictly to verified campus students.

4. 3-layer fraud control :

   Verify owners using hidden detail questions, digital OTP handover codes and designated CCTV safe zones.

5. Authentication and security:

   Provide security and user authentication through verified campus email and OTP handover validation.

6. Lifecycle audit and incentives:

   Capture exact discovery/return logs and calculate a finder reward and platform commission split.

7. Guest proxy support:

   Allow non email campus visitors to log items via security staff using a Flutter admin panel.

4. Technical Architecture & stack:

    1. Frontend:

       Flutter (Mobile app)& Flutter web(admin panel).

    2. Backend API:

       Node.js/Express.js.

    3. Database:

       Mongo DB Atlas(M0 free cluster).

    4. AI engine:

       Google ML Kit (image labeling SDK on device).

    5. Notifications:

       Firebase cloud Messaging (**FCM).**

5. High level database schema:

   Each lost and found item record is stored in **Mongo DB Atlas** as a unified JSON document.

   This structure maintains a complete life cycle audit trail including discovery details, return logs and incentive calculations .

6. Timeline:

   Phase1:

   (Week 1-4)

   Mongo DB schema

   Node.js API setup

   @iub.edu.pk authentication engine &UI

   Wire Frames.

   Phase 2:

   (Week 5-8)

Google ML Kit integration

Weighted Matching logic

FCM alerts & In-apo chats

Phase 3

(Week 9-12)

OTP verification engine

Reward logic

Flutter web admin panel

Testing & Documentation

7. Financial Feasibility

Development cost: 0PKR

(Open source flutter framework)

Database & server Hosting:0PKR

(MongoDB Atlas & Render free tires)

AI & Notifications fee: 0PKR

(Google ML Kit on-device &FCM free push messaging)

Total project cost: 0PKR

(100%free and self sustaining)